ESTACIÓN DE MONITOREO AMBIENTAL — ESTRUCTURA DEL ESQUEMÁTICO

Alimentacion

File: power.kicad_sch

MCU

File: mcu.kicad_sch

Perifericos

File: io.kicad_sch

Bloques:

1) Alimentación: USB-C (5 V) -> TP4056 -> celda 18650 1S protegida (DW01A + FS8205A).

   VSYS = VBUS (vía SS14) o BAT+ (vía AO3401A). VSYS alimenta el LDO AP2112K-3.3 y el puente H.

2) MCU: ESP32-WROOM-32E, programación por CH340C con reset automático DTR/RTS.

3) Periféricos: L9110S + motorreductor, AHT20 (I2C), FC-51 (IR), reed, finales de carrera, LEDs y buzzer.

Todas las conexiones entre hojas se realizan por etiquetas globales con el mismo nombre de red.

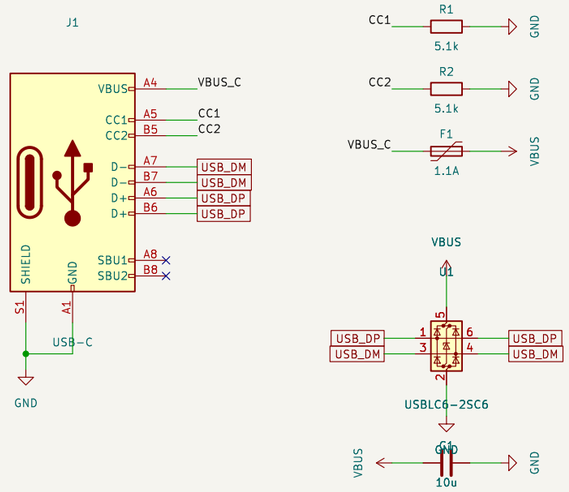
Hoja raíz — estructura jerárquica
J. Ropero, S. Rueda
Universidad Autónoma de Manizales — Diseño Electrónico CAD

Sheet: /
File: estacion.kicad_sch

Title: Estación de Monitoreo Ambiental con Ventilación Automática

Size: A3	Date: 2026-11-04	**Rev: v2.0**
KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4	Id: 1/4

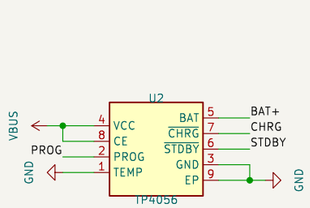
ENTRADA USB-C

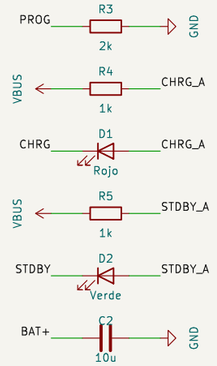


CELDA Li-Ion 1S

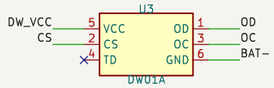


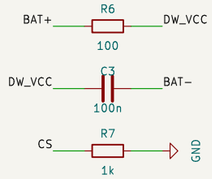
CARGADOR TP4056 (Ichg = 1200/R3 = 600 mA)

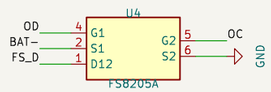




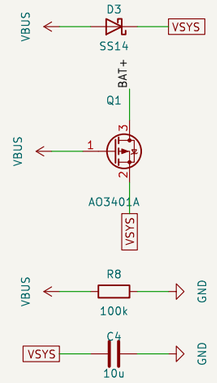
PROTECCIÓN DE CELDA DW01A + FS8205A



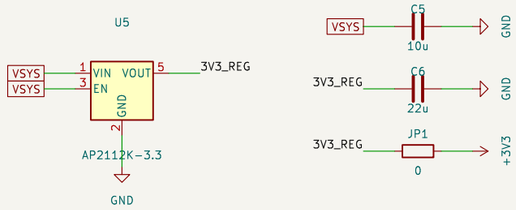




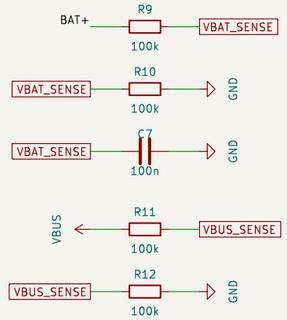
RUTA DE POTENCIA VBUS / BATERÍA



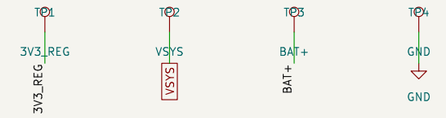
REGULADOR LDO 3.3 V + PUENTE JP1 (REQ-DFB-001)



MEDICIÓN VBAT / VBUS (ADC1)



PUNTOS DE PRUEBA (REQ-DFB-002)



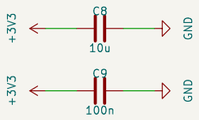
BANDERAS ERC



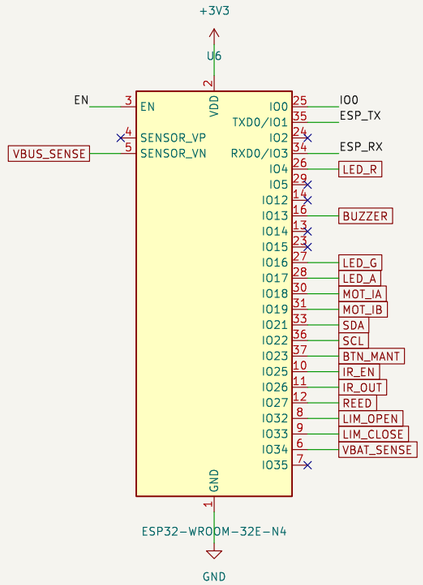
Alimentación: USB-C, cargador, protección y regulador 3.3 V
J. Ropero, S. Rueda
Universidad Autónoma de Manizales – Diseño Electrónico CAD

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KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 2/4

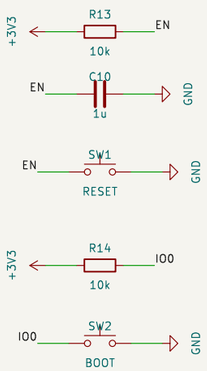
DESACOPLE 3.3 V



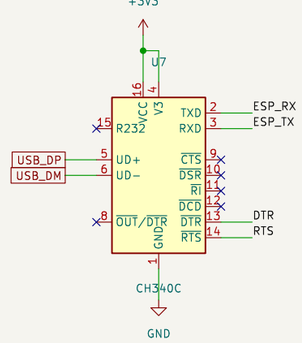
MÓDULO ESP32-WROOM-32E



RESET / BOOT

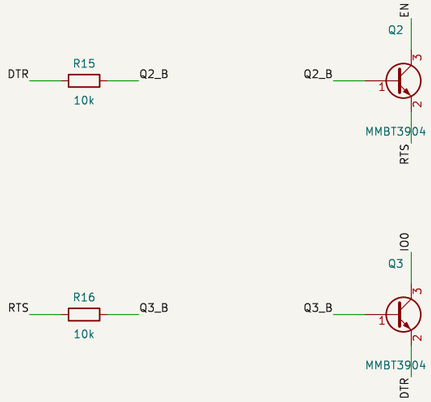


CONVERSOR USB-UART CH340C





RESET AUTOMÁTICO DTR/RTS



PUNTOS DE PRUEBA



ESP32-WROOM-32E y programación USB-UART
J. Ropero, S. Rueda
Universidad Autónoma de Manizales – Diseño Electrónico CAD

Sheet: /MCU/
File: mcu.kicad_sch

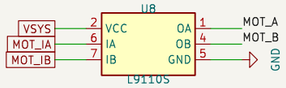
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PUENTE H L9110S (alimentado desde VSYS, REQ-PER-002)

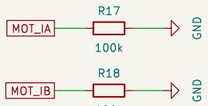






MOTORREDUCTOR

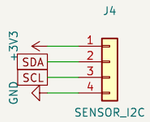


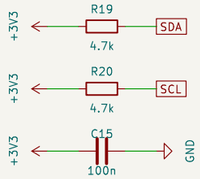


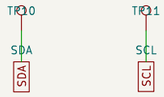




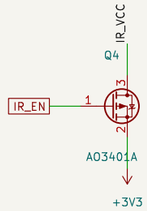
SENSOR AMBIENTAL I2C (AHT20)

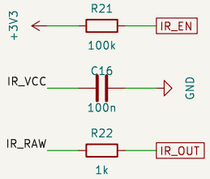


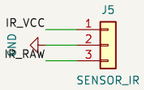




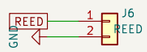
SENSOR IR FC-51 CON ALIMENTACIÓN CONMUTADA

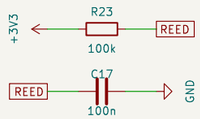


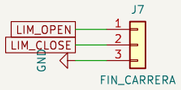


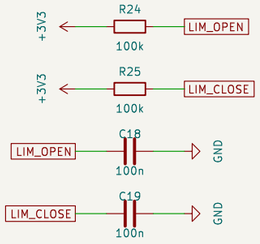


SENSOR MAGNÉTICO / FINALES DE CARRERA

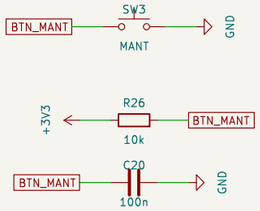








BOTÓN MANTENIMIENTO

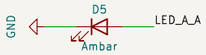


LEDs DE ESTADO (REQ-FUN-006)





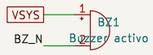




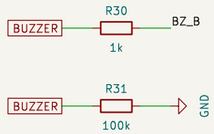


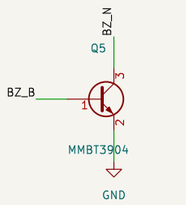


BUZZER DE ALARMA









ORIFICIOS DE MONTAJE



Actuación, sensores e interfaz de usuario
J. Ropero, S. Rueda
Universidad Autónoma de Manizales – Diseño Electrónico CAD

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